

Problem #5: Making Sense of Turkish Words (20 marks)

Solution:

Turkish is a highly *agglutinative language*, which means that the root words can be joined with each other as well as with prefixes and suffixes to form new words. By looking at the common letter sequences within the given set of Turkish words, we can deduce the various roots and suffixes of Turkish and come up with the morphological decomposition of the words. By comparing the semantic relations between the English words to the morphological decomposition of the Turkish words, we can match the words in the two languages. Note that Turkish words undergo *vowel assimilation*. So a suffix, say *ci*, can occur as *ci* when the previous vowel is *e* or *i* (as in **gözlemci** or **şarkici**) or *cü* if the previous vowel is *ü* (as in **gözlükcü**).

Assignment 1:

gözlemci = **göz** + **lem** + **ci** = eye + abstract-noun marker + suffix for “doer” = **observer**

gözlükçü = **göz** + **lük** + **cü** = eye + state-of-being + suffix for doer = spectacles + suffix for doer = **optician**

gözlemek = **göz** + **lem** + **mek** = eye + abstract-noun marker + verb marker = **to observe**

döndürmek = **döndür** + **mek** = rotation + verb marker = **to rotate**

güneşli = **gün** + **eşli** = sun + adjective marker = **sunny**

gündöndü = **gün** + **döndü** = sun + rotation = That which rotates with the sun = **sunflower**

şarkici = **şarki** + **ci** = song + suffix for “doer” = **singer**

çocukluk = **çocuk** + **luk** = child + state-of-being = **childhood**

Pazar = **Sunday**

Pazartesi = **Pazar** + **tesi** = Sunday + next-day = Day after Sunday = **Monday**

Cumartesi = **Cumar** + **tesi** = Friday + next-day = Day after Friday = **Saturday**

Assignment 2:

observation = **gözlem**

child = **çocuk**

the state of being a singer = **şarkicilik**

spectacles = **gözlük**

Friday = **Cumar** (Actually it is **Cuma**, but from the given data one cannot deduce it)
